

Design and Prototype 2 – Polyline Editor

1. Introduction

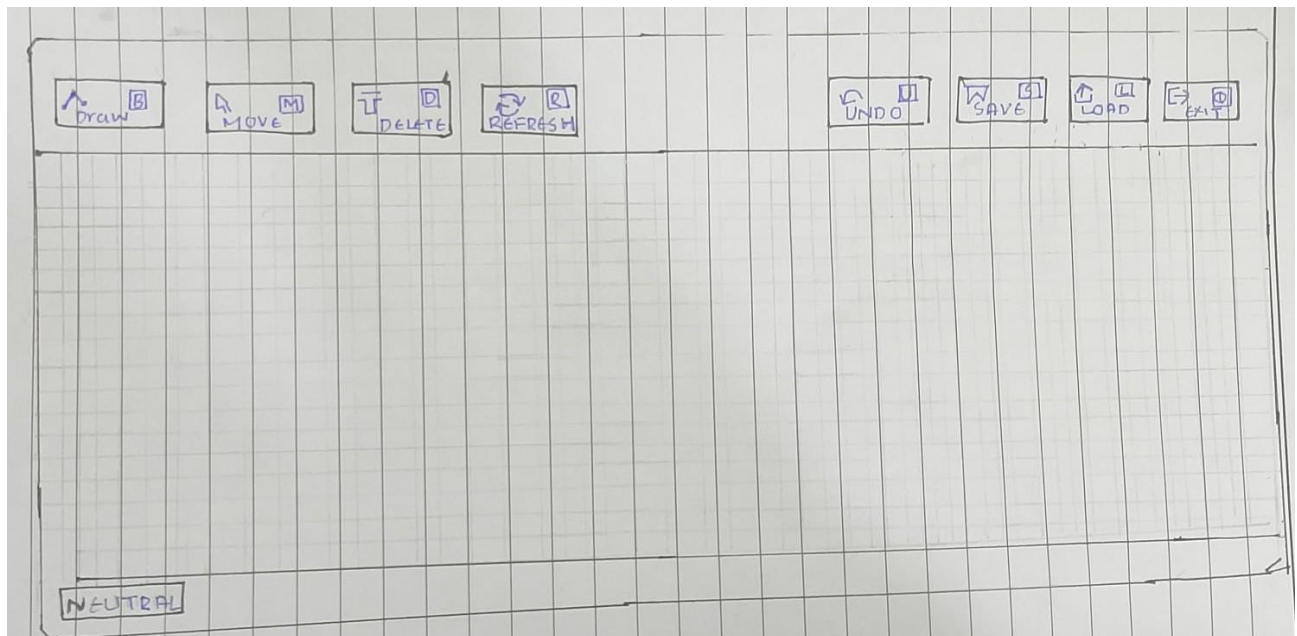
This document presents the design and improved Prototype 2 of the Polyline Editor. It builds upon the issues identified in Prototype 1 and introduces new features and interaction improvements to enhance usability, efficiency, and user control

2. Design Goals

The main goals of Prototype 2 are:

- Improve user control and flexibility
- Reduce errors and support recovery
- Enhance interaction through better feedback
- Increase functionality of the editor
- Make the system more intuitive and user-friendly

3. Low-Fidelity Design (Prototype 2)



4. New Features Introduced in Prototype 2

Undo Function

Allows users to reverse their last actions up to three times.

Lock Feature

Enables users to lock specific polylines to prevent editing.

Save/Load Functionality

Allows users to save their work and reload it later.

This ensures data persistence and supports long-term usage of the system.

Neutral State

The system now includes a neutral state where no specific action (draw, move, delete) is active.

Improved Feedback

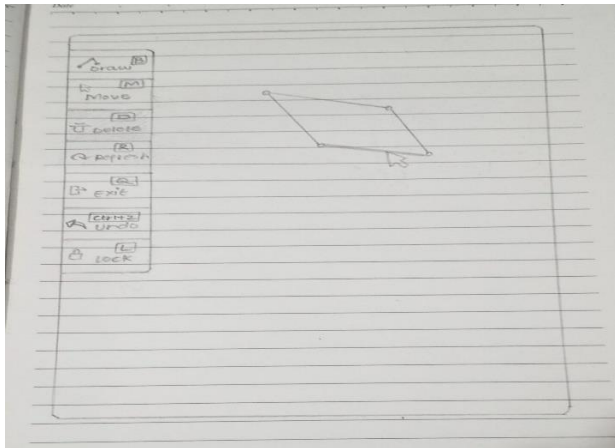
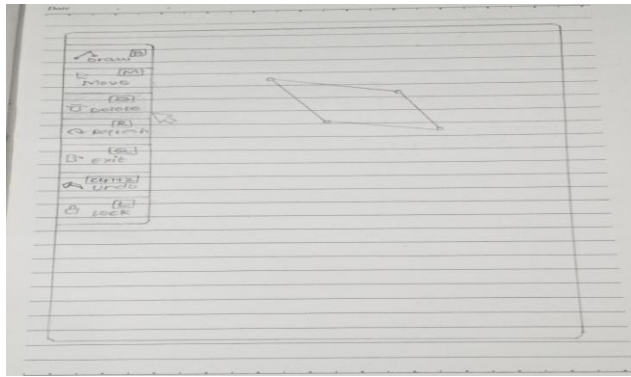
System now shows messages after actions.

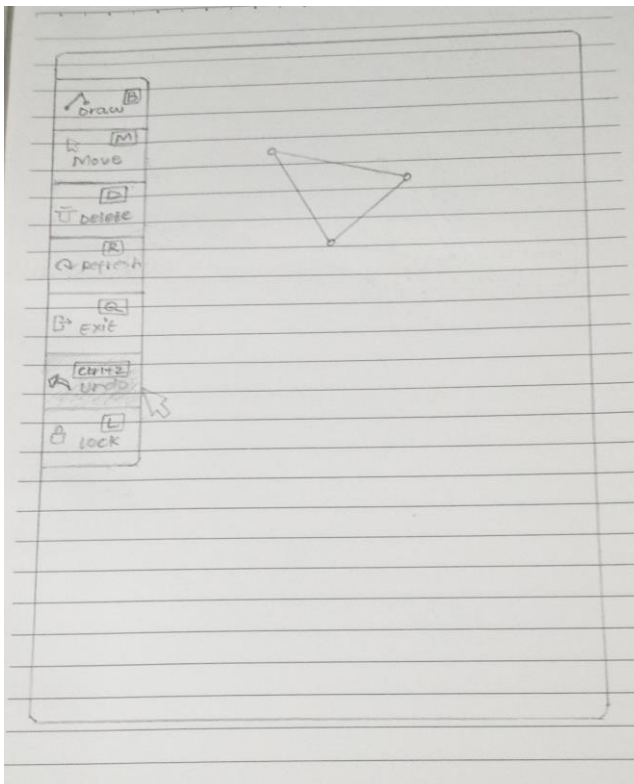
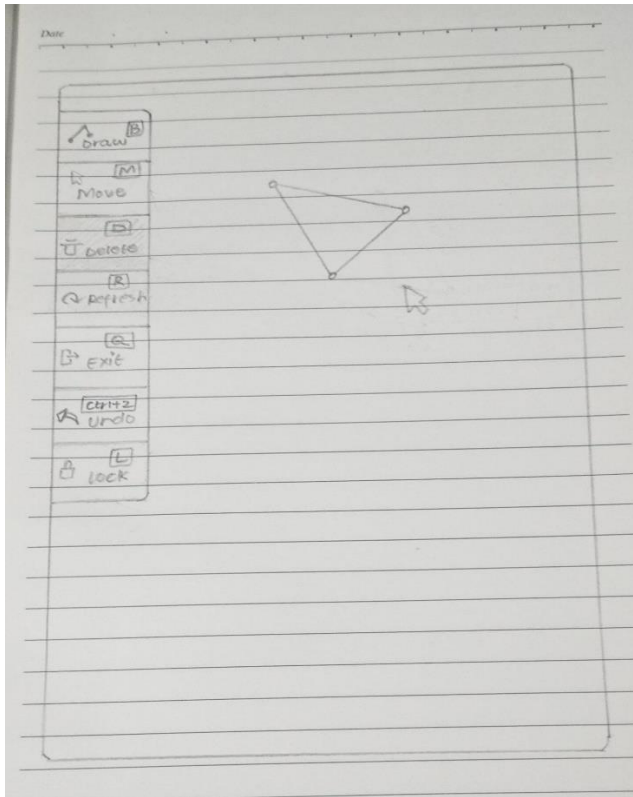
Visual Highlighting

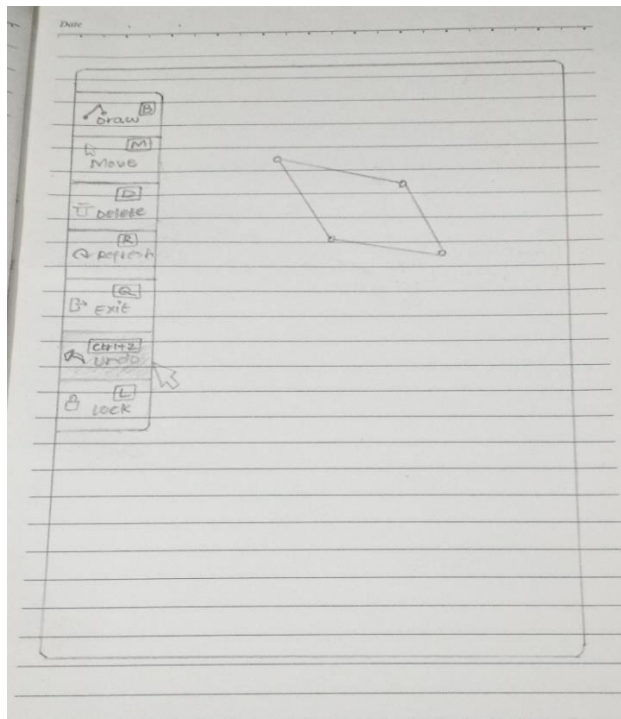
Selected points are highlighted for better visibility.

5. Mapping User Actions To New Functions

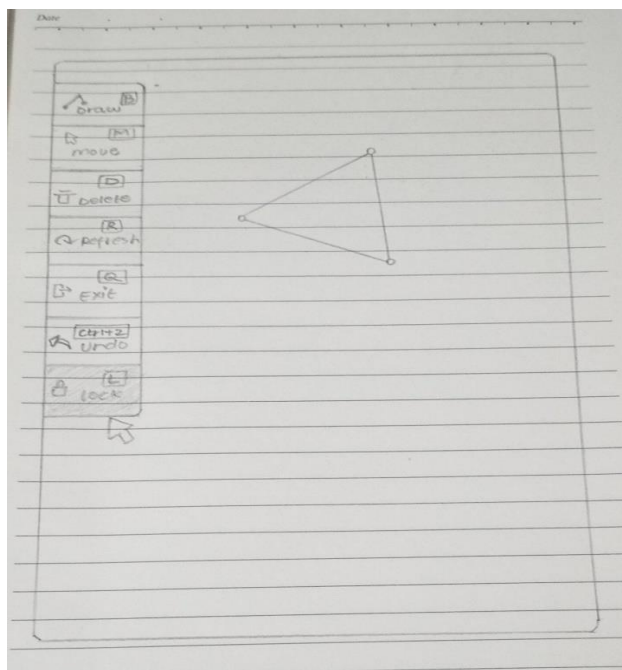
- **Undo Function**

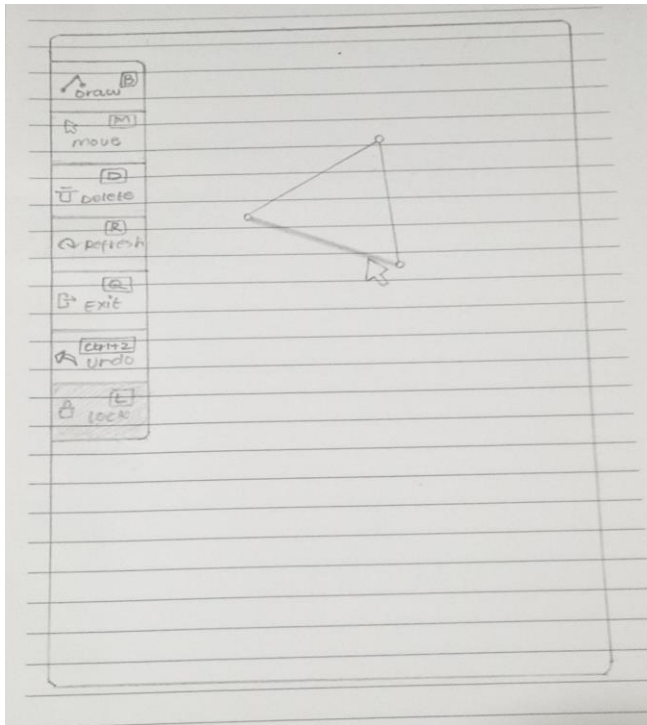


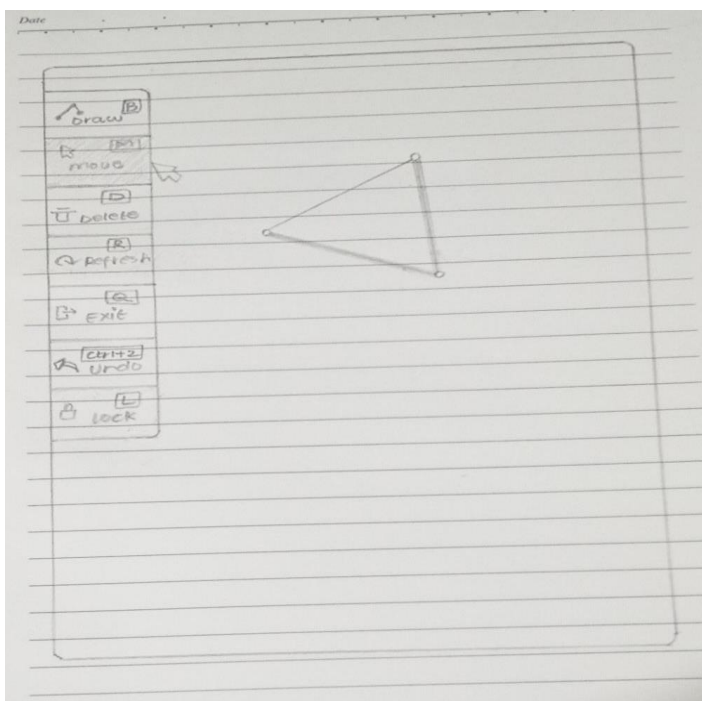
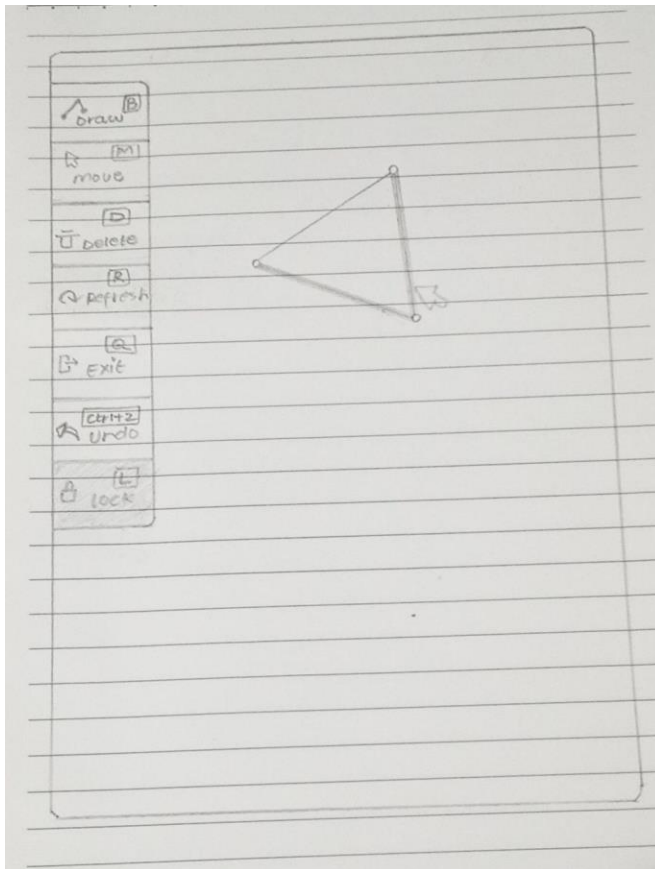


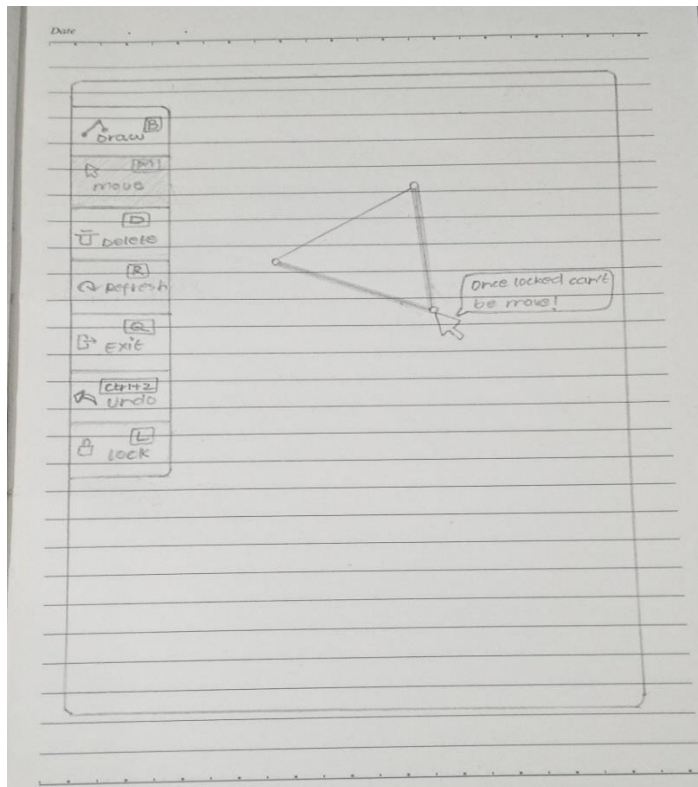


- **Lock Function**



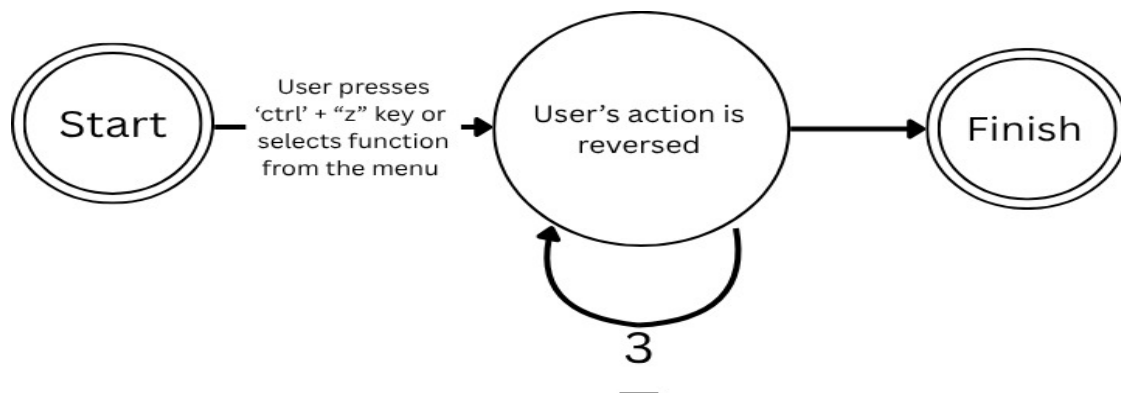




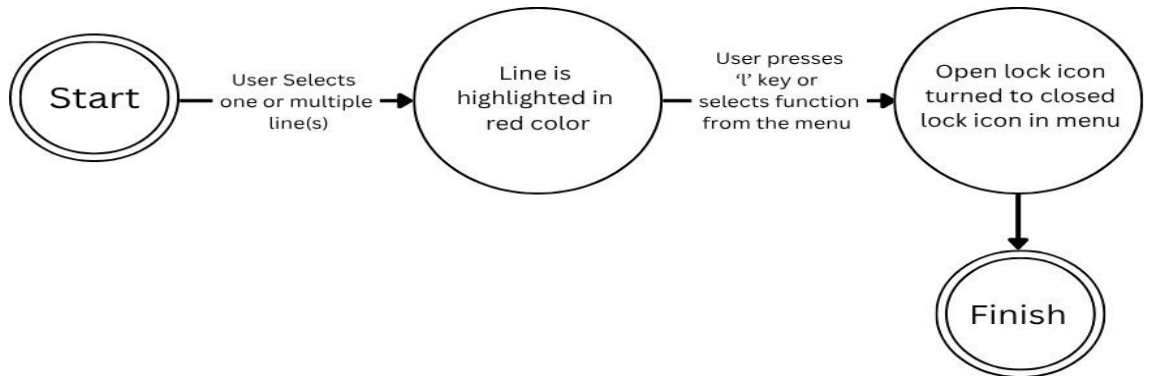


6.STATE TRANSITION NETWORK OF NEW FUNCTIONS

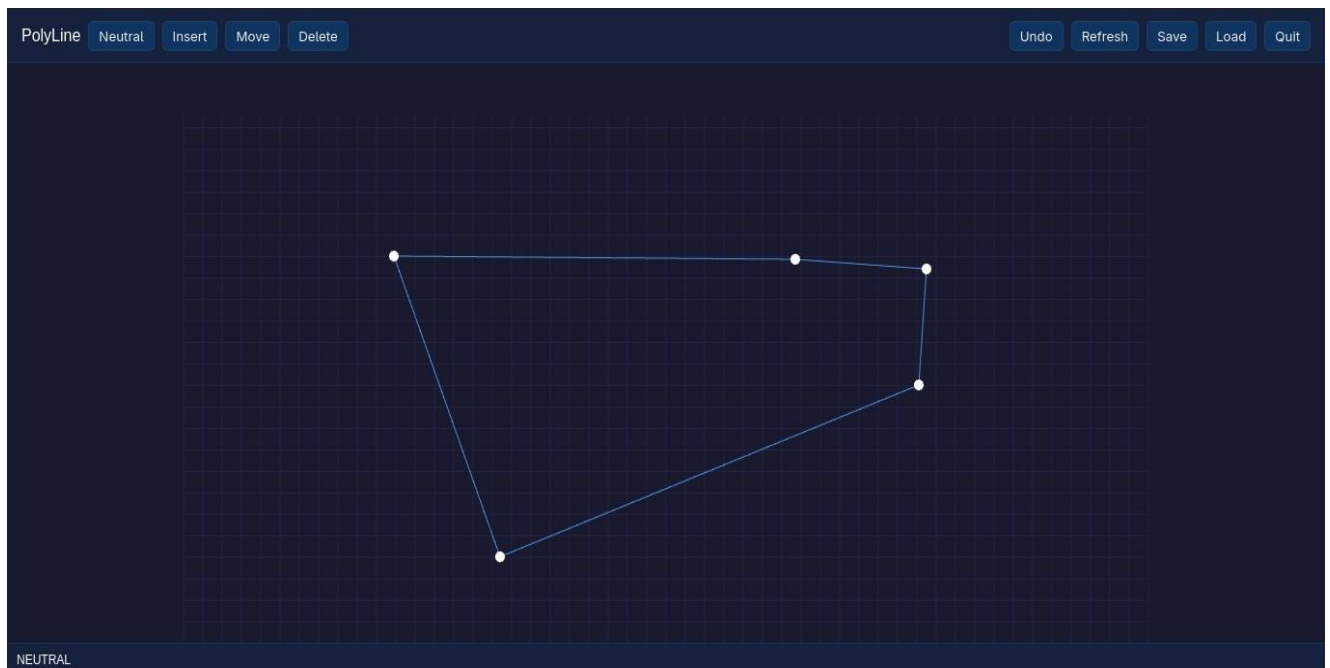
- UNDO



- **LOCK**



7. Prototype Implementation (Version 2)



Prototype 2 Analysis – Polyline Editor

1. Introduction

This section evaluates Prototype 2 of the Polyline Editor, focusing on improvements made over Prototype 1. The analysis examines usability, user interaction, system feedback, and overall effectiveness based on Human–Computer Interaction (HCI) principles.

2. Evaluation Method

The prototype was evaluated by performing key user tasks such as drawing, moving, deleting, saving, loading, and undoing actions. These tasks were analyzed to observe improvements in usability, efficiency, and error handling. The system was also compared against Prototype 1 to assess how previously identified issues have been addressed.

3. Usability Improvements Over Prototype 1

Prototype 2 successfully resolves many of the issues identified in Prototype 1:

Mode Visibility Improved

The introduction of a neutral state and clearer interaction flow reduces confusion about system behavior.

Enhanced User Feedback

The system now provides messages after each action, improving user awareness and interaction clarity.

Error Recovery Introduced

The undo feature allows users to reverse actions, reducing frustration and increasing confidence.

- **Improved Functionality**
Additional features such as save/load and lock significantly enhance system capability.
- **Better Visual Interaction**
Highlighting of selected points improves visibility and ease of interaction.
- **Data Persistence Achieved**
Save and load functionality allows users to preserve and reuse their work.

5. HCI Principles Achieved

Prototype 2 better aligns with key HCI principles:

Feedback

The system provides immediate responses after user actions.

Visibility of System Status

Users are informed about system state through messages and interaction behavior.

Error Prevention

Neutral state and lock feature minimize accidental actions.

Error Recovery

Undo functionality allows correction of mistakes.

Consistency

Interaction methods remain uniform across different operations.

User Control and Freedom

Users can easily manage actions and reverse them when needed.

6. Remaining Limitations

Despite improvements, some limitations still exist:

- Undo functionality limited to three steps
- Save/load may not support advanced file formats
- Performance may decrease with complex or large drawings

7. Conclusion

Prototype 2 shows significant improvement over Prototype 1 in terms of usability, functionality, and interaction design. The addition of features such as undo, save/load, lock, and neutral state enhances user control, reduces errors, and improves overall system efficiency. While some limitations remain, the prototype demonstrates a strong application of HCI principles and provides a more effective and user-friendly interaction experience.